Testing shock function alternatives for stock prediction models using GBM in MS Excel

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1 Introduction

This report contains the results of an investigation conducted into how the price of a stock is affected by a mixture of drift and random shocks amplified by volatility. Different types of shock functions involving $\ln$ and $\tan$ functions are considered, where the stock prices of blue-chip stocks are predicted using Geometric Brownian Motion and two new shock functions. This was done for 11 days ahead of the first day, and was used with a Monte Carlo simulation of 500 runs each to analyse the mean and standard deviation of percentage error between the predictions of those models.

If you only want instructions on how to read and use the Excel file, go to Section 9: Instructions for use and layout.

2 Factors affecting a stock price

This report will focus mainly on the stock price over mid-to-high frequency timescales for trading, and will not account for other factors that are predictable, such as the slow growth-recession cycle of economics and their effects on prices.

The two main factors affecting a stock price are stock drift and random shocks. Stock drift is caused by a variety of factors, is stable, slow and predictable (eg: linear or periodic), and represents the "average" direction of movement. Random shocks are proportional to stock volatility, caused by relevant news and changes in other asset classes, and are modelled using stochastic calculus and differential equations.

By far the most common shock function used across stock price prediction algorithms is the standard normal inverse function (referred to as norm.s.inv). If you assume that stock returns are normally distributed (a common and generally untrue assumption), then you can interpret it as a probability density function (p.d.f.), and if you take its cumulative distribution, you obtain a function that resembles a flattened 'S'. Taking the inverse of this non-elementary function, you see that it also resembles a

$\tan$ graph, which interestingly enough is the inverse of the c.d.f. of a Lorentz distribution (which will be shown later). *Note*: In my research paper published with Parabola (linked at the end of this section), I addressed the assumption of normally distributed returns, and derived an alternative Gram-Charlier p.d.f, which can model asymmetrical markets more accurately but plotting or using an inverse c.d.f with it is currently a future avenue of research.

For this report, it is assumed that the probability of a shock function increasing or decreasing the stock price at a given time is 0.5, and that all other factors that do not conform to this are accounted for in the volatility term (this is not true but it simplifies my work a great deal, so please bear with it).

Link to the paper: Weighting for a better distribution - Gram-Charlier in the world of finance.

3 Variants of the standard normal inverse

Note that x is given by rand$(0, 1)$.

In Excel, which is what the investigation was conducted with, this function can be accessed using `norm.s.inv()`, where the argument belongs to $[0, 1]$. Using Desmos (and a few clever tricks) it is possible to graph this function, which is seen in Figure 1:

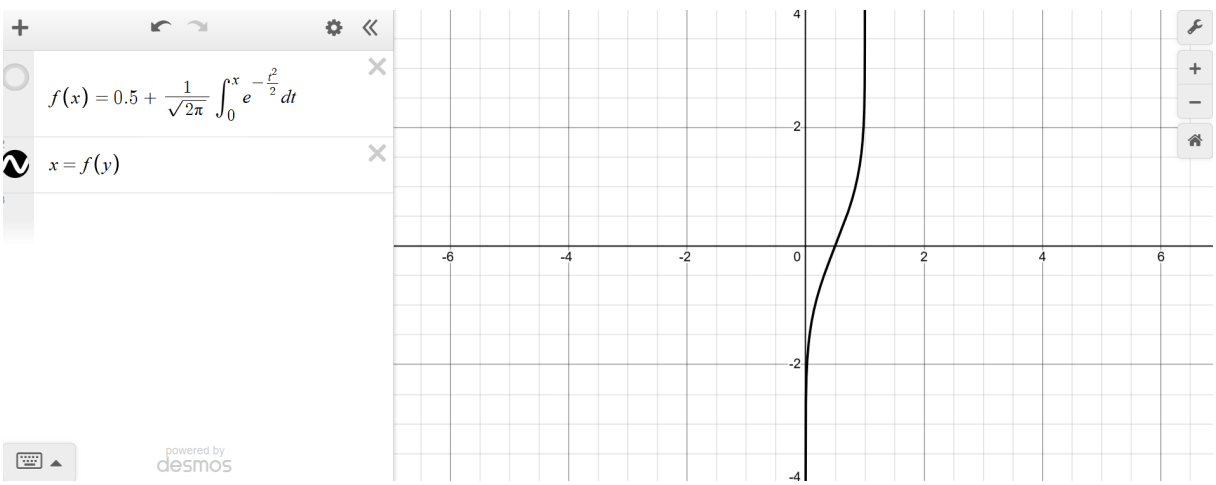

Figure 1: Standard Normal Inverse graph $y = \Phi^{-1}(x)$

However, a large family of even p.d.f. functions' inverses produce this approximate shape. Consider the graph (which was obtained using trial and error):

$$f(x) = \frac{e^x}{(e^x + 1)^2}\,.$$

This function is even, since

$$f(-x) = \frac{e^{-x}}{(e^{-x}+1)^2} \times \frac{e^{2x}}{e^{2x}} = \frac{e^x}{(e^x+1)^2} = f(x)\,.$$

, so it satisfies the assumption that the stock price has an even chance of going up or down at any time. Consider the following integral, which, if finite, would confirm that it is able to be turned into a valid p.d.f.:

$$\int_{-\infty}^{\infty} \frac{e^x}{(e^x+1)^2}\,dx = \int_0^{\infty} \frac{1}{(v+1)^2}\,dv = \lim_{\alpha\to\infty} \int_0^{\alpha} \frac{1}{(v+1)^2}\,dv\,,$$

where the substitution $u = e^x$, $\ln(u) = x$ was used, with $x : -\infty \to \infty$ mapping to $u : 0 \to \infty$. Continuing,

$$= \lim_{\alpha\to\infty} \left[\frac{-1}{v+1}\right]_0^{\alpha} = \lim_{\alpha\to\infty} \frac{-1}{\alpha+1} + \frac{1}{1} = \lim_{\alpha\to\infty} 1 - \frac{1}{\alpha+1} = 1\,.$$

Therefore $\dfrac{e^x}{(e^x+1)^2}$ is a valid p.d.f.

By modifying the bounds and changing ∞ to x, you can find the c.d.f. $F(x)$:

$$F(x) = \int_{-\infty}^{x} \frac{e^t}{(e^t+1)^2}\,dt\,,$$

which can be integrated using the substitution above to give

$$F(x) = \frac{e^x}{e^x+1}\,.$$

This is a valid c.d.f. as $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$.

Now you can try to find its inverse, making its new domain $[0, 1]$:

$$y = \frac{e^x}{e^x+1} = \frac{1}{1+e^{-x}}\,,$$

so that

$$\begin{aligned}
x &= \frac{1}{1+e^{-y}}\,,\\
1+e^{-y} &= \frac{1}{x}\,,\\
e^{-y} &= \frac{1}{x} - 1 = \frac{1-x}{x}\,,\\
-y &= \ln\left(\frac{1-x}{x}\right)\,,\\
y &= \ln\left(\frac{x}{1-x}\right)\,.
\end{aligned}$$

The new second function $y = \ln\left(\frac{x}{1-x}\right)$ has a graph like in Figure 2:

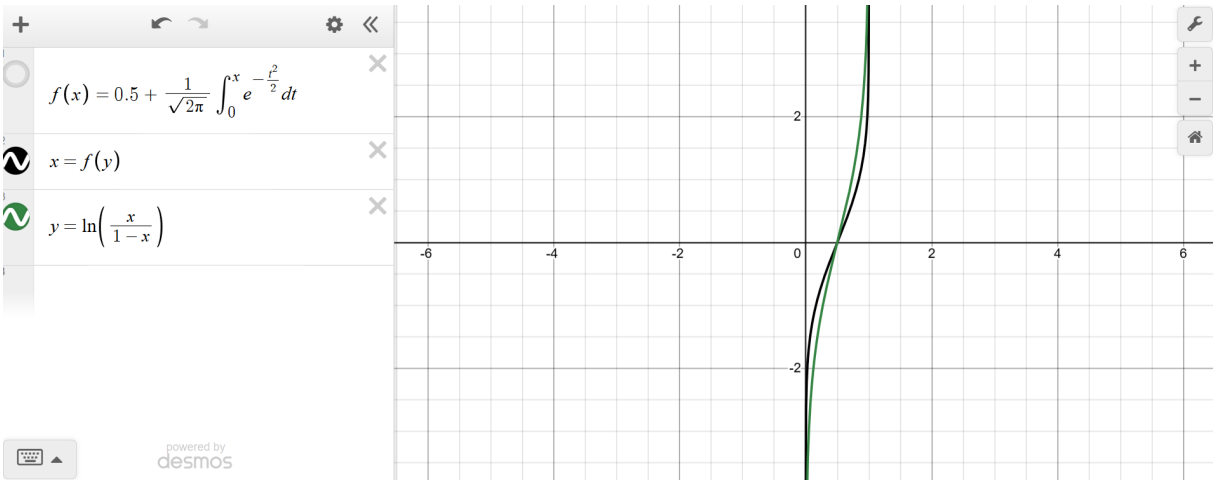

Figure 2: $\ln$ graph (green) and norm.s.inv (black).

This closely resembles the standard normal inverse function and so you can try to use it as a replacement for the random shock function. We can observe here that the graph, at its x intercept is steeper than the norm.s.inv curve, and that the magnitude of the values of the ln graph is greater for majority of the interval [0,1], which leads us to hypothesise that our shocks would be more extreme than those seen from the norm.s.inv curve.

Another graph that looks strikingly similar to this shape, with asymptotes at $-\pi/2$ and $\pi/2$, is the $\tan$ graph. Using simple graph transformations, you can modify the basic $\tan$ into a one with a modified domain $[0, 1]$ as follows:

$$f(x) = \tan\left(\pi x - \frac{\pi}{2}\right), \quad x \in [0, 1].$$

By restricting the domain, you shrinks it to match the range of outputs provided by `rand()` in Excel. As will be seen in testing, this function is incredibly varied, and can occasionally produce outputs up to a million percent error. To combat this, you can multiply it by a term much smaller than 1, which increases its "band of stability" by reducing the values around the middle of the interval. This term is called γ.

The $\tan$ function was added purely because its inverse's derivative resembles the Lorentz distribution, as seen below. Starting from

$$y = \gamma \tan\left(\pi x - \frac{\pi}{2}\right),$$

you obtain

$$\frac{y}{\gamma} = \tan\left(\pi\left(x - \tfrac{1}{2}\right)\right),$$

$$\frac{1}{\pi}\tan^{-1}\left(\frac{y}{\gamma}\right) = x - \frac{1}{2},$$

$$x = \frac{1}{\pi}\tan^{-1}\left(\frac{y}{\gamma}\right) + \frac{1}{2}\,.$$

Therefore the inverse function is:

$$y = \frac{1}{\pi}\tan^{-1}\left(\frac{x}{\gamma}\right) + \frac{1}{2}\,.$$

Taking the derivative and letting $u = \dfrac{x}{\gamma}$ (so that $\dfrac{du}{dx} = \dfrac{1}{\gamma}$):

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}\,.$$

Now

$$\frac{dy}{du} = \frac{1}{\pi} \times \frac{1}{1+u^2} = \frac{1}{\pi(1+u^2)}\,,$$

so

$$\frac{dy}{dx} = \frac{1}{\pi\gamma\left(1+\dfrac{x^2}{\gamma^2}\right)} = \frac{1}{\pi\left(\gamma+\dfrac{x^2}{\gamma}\right)} = \frac{\gamma}{\pi(\gamma^2+x^2)}\,,$$

which resembles the Lorentz distribution. It is also a valid distribution, since

$$\int_{-\infty}^{\infty} \frac{1}{\gamma^2+x^2}\,dx = \frac{\pi}{\gamma}\,,$$

so

$$\frac{\gamma}{\pi}\int_{-\infty}^{\infty} \frac{1}{\gamma^2+x^2}\,dx = 1\,.$$

Therefore $\dfrac{\gamma}{\pi(\gamma^2+x^2)}$ is a valid p.d.f., making the inverse c.d.f. $y = \gamma\tan\left(\pi x - \dfrac{\pi}{2}\right)$ a valid inverse function. It is graphed below in Figure 3 against norm.s.inv :

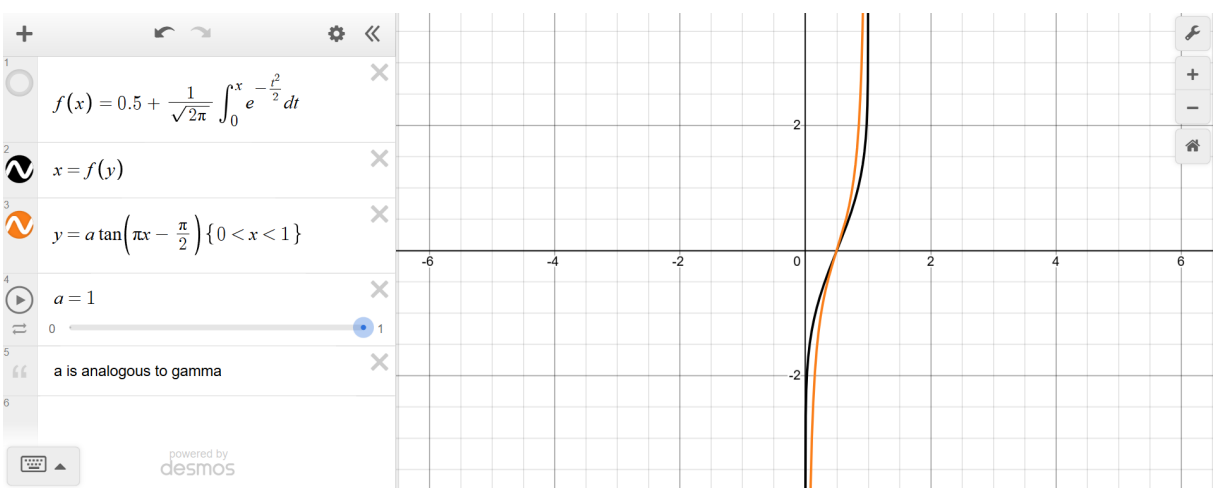

Figure 3: Tan graph(orange) and norm.s.inv (black)

Hence, there are now 2 more "lookalike" functions in place of the standard normal distribution for the shock function:

$$y = \ln\left(\frac{x}{1-x}\right)$$

and

$$y = \gamma \tan\left(\pi x - \frac{\pi}{2}\right), \quad x \in [0,1]\,.$$

4 A quick introduction to Geometric Brownian Motion

Geometric Brownian Motion (GBM) is a solution to a stochastic differential equation governing random, erratic movement (like volatility in stock prices). The differential equation is:

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t\,,$$

where $dW_t \sim$ dist$(0, dt)$ (generally assumed to be $N(0,1)$, and in our case, it is the p.d.f behind the shock function).

When discretised, this becomes:

$$S_{t+1} = S_t \cdot \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\, Z\right],$$

where Z is the shock function of a random number in$[0,1]$.

The GBM is the simplest model for modelling stock price volatility, and while it makes assumptions, those assumptions help you understand the effects of just the shock function on the prices of the stock, ignoring the effects of all other features of other models (for example, the Cox–Ingersoll–Ross model reduces the size of volatility as price approaches 0, keeping prices positive). This allows you to objectively assess the performance of different stocks independently and distinguish the characteristics caused by the shock function from the model.

5 Analysis method

In the model (using GBM prices), the stock price was predicted iteratively up to 11 days ahead, after which the absolute error as a percentage was measured using the following formula:

$$\text{Hist.Delta\%} = \frac{|\,\text{Predicted price} - \text{Historical price}\,|}{\text{Historical price}} \times 100\%\,.$$

This was then done 500 times using a Monte Carlo simulation for each shock function, after which the mean and standard deviation of each of the 500 runs was found and graphed. This process was done for MSFT and TSLA prices just to see how it performs with blue-chip stocks, but it can be done with any other kind.

6 Results Summary

Note: some values may change upon spreadsheet refreshing.

6.1 Standard normal inverse ($y = \Phi^{-1}(x)$):

- For MSFT, mean error $\approx 3.5\%$, std. deviation $\approx 2.0\%$.
- For TSLA, mean error $\approx 7.5\%$, std. deviation $\approx 4.5\%$.
- Indicates that the model is not great at prediction, but can be used to give the approximate ballpark of the price.
- Pretty average performance for a shock function.
- Used normally for Monte-Carlo simulations using Python or C++ in quantitative trading firms due to ease of use and high optimisation.

6.2 Logarithm function $\left(y = \ln\left(\frac{x}{1-x}\right)\right)$:

- For MSFT, mean error $\approx 3.5\%$, std. deviation $\approx 1.4\%$.
- For TSLA, mean error $\approx 8.3\%$, std. deviation $\approx 3.2\%$.
- Model has approximately the same metrics as the standard normal inverse. Tends to have a slightly higher margin of error while having a smaller standard deviation.
- Tends to be wrong more often (above mean), but is quite consistent at it (below std. deviation) compared to norm.s.inv.

6.3 Tangent function $\left(y = \gamma \tan\left(\pi x - \frac{\pi}{2}\right)\right)$:

- Values for both can swing and vary wildly with every refresh of the spreadsheet.
- These are its values when it is stable:
  - For MSFT, mean error ≈ 2.5–2.7%, std. deviation ≈ 1.0–8.5%.
  - For TSLA, mean error $\approx 2.5\%$, std. deviation ≈ 2.0–5.0%.
- When it tends to be most unstable (which is about half the time), its values for mean and standard deviation can be anywhere between 100% to $5 \times 10^6\%$.
- This extreme variation makes it highly unreliable to use as a shock function, due to its large margins of error when incorrect.

7 Conclusion

- The standard normal inverse is better for certain stocks, but it can be interchanged with the logarithm variant as needed.
- The $\tan$ function is extremely unstable, even though its shape closely matches that of the $\ln$ and standard normal inverse curves.
- This may be due to the excess kurtosis that is present in a Lorentz distribution compared to a normal distribution.

8 Discussion

A few notable things to bring to attention for the model are the discrepancies I was unable to explain in the functions' behaviour so far.

The first of which was the general trend of the $\ln$ function having a smaller mean error than the norm.s.inv, which seems contradictory because the former is more steep and it's function values are greater in magnitude, which would imply that the shock function is more powerful, but the standard deviation for both blue-chips' prices are lesser when $\ln$ was used compared to norm.s.inv.

The second was the incredibly high variance of the $\tan$ function. The general shape of the function closely resembles that of the norm.s.inv, albeit more extreme, but it is significantly more prone to producing error percentages that are multiple degrees of magnitude greater than the ones produced by the other two functions, and if γ was set to 1, then it is almost guaranteed.

However, if we greatly decrease the value of γ (to approximately 0.005), we see that these anomalies' frequency is greatly reduced, even though the shape of the graph tends to diverge a lot from the initial norm.s.inv function. I currently have no explanation for this, and any ideas explaining why are more than welcome.

Another seemingly interesting thing to note is that if you set the value of γ to be equal to 0, the 'random' shock aspect of the prediction is fully cancelled out, and doing so actually tends to produce a smaller mean error percentage across the Monte Carlo simulation (than if a shock function was present) and causes standard deviation to plummet to 0. This implies that assuming a stock remains unchanged if you are unsure of its direction of movement is most likely your best bet, as your error is marginally lower.

9 Instructions for use and layout

All necessary inputs are yellow cells with red text. For example, A1 is the ticker of the stock that you want to analyse, and U1 is the value of γ in the $\tan$ expression.

Instructions for use

(a) Simply input the ticker of the stock you wish to analyse in cell A1, and change the value in U1 if required. Any stocks on the NYSE will be accepted if you just enter their ticker, but for stocks not on the NYSE, you need to add the exchange code, followed by a colon, then the ticker (Eg: XASX:BHP). Some common exchange codes include XNYS (default, New York Stock Exchange), or XLON (London Stock Exchange).

(b) The initial values of the graphs should all be 0 when the ticker is changed. This is normal. Head to Formulas → Calculation and click "Calculate Sheet".

(c) Then the 500 Monte Carlo simulations will be rerun and the graphs will update as required.

(d) If the standard deviation or mean error for the $\tan$ function is very high, this is normal and it is a characteristic of the function. Just recalculate the sheet again and it will work.

Layout

- The top 2 rows are for labelling, headers and inputs.
- Cell A2 holds the `stockhistory()` function, which pulls historical data from the Microsoft database on a daily interval.
- This data spills out across from $A1 - B504$, with $C3 - C504$ containing the log returns used for GBM. (Note that these are *NOT* the log of the absolute standard return and are calculated as $\ln(\frac{S_t}{S_{t-1}})$.)
- The return metrics are housed from $E1 - F12$, which are used in the calculations for the GBM.
- The trials of the shock functions and predictions for the next 11 days are between $K1 - V13$, with the historical data of the same period from $H1 - I13$.
- The 500 Monte Carlo simulations for the 3 functions are from $Z1 - AG502$.
- The simulation summary tables as well as the error mean and standard deviation graphs are between $D15 - P33$.

IMPORTANT NOTICE:

Keep in mind that the `stockhistory()` function has to be working on your version of Excel and your associated Microsoft account, or else the data will not load/be blocked and the model, graphs and simulations will not work at all.